Probability

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Lecture 1

Abstract

3 Axioms:

– Sample Space

– Σ-Algebra

– Probability

5 Concepts:

– Random Variables

– Distribution Function

– Density

– Moment（矩）

– Characteristic Function Generating

2 Relations:

– Independence

– Conditional

6 Distributions:

– Bernoulli

– Uniform（均匀分布）

– Binomial（二项分布）

– Exponential（指数分布）

– Gaussian

– Poisson

4 Calculations:

– Probability

– Distribution

– Expectation

– Variance（方差）

3 Skills:

– Linearity → Expectation

– Conditional Expectation → Expectation

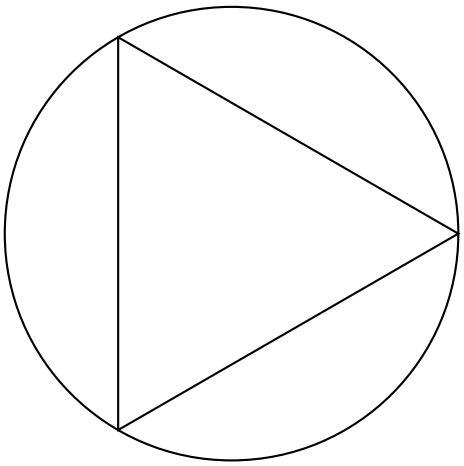
– Multivariate Integral

Lecture 2

Part 1

Uncertainty → Statistical Experiment → Sample Space Ω → Subsets

e.g. Q: "Randomly choose chord", $P_{\text{chord}\geq\text{triangle's side}}$?



Key Point: What is the Sample Space?

- {points on the circular arc}: 1/3
- {points in the circle}: 1/4

Part 2

P: $\Sigma \to [0,1]$, Domain: set of sets

Property:

- (1) $P(\Omega) = 1$, $P(\emptyset) = 0$
- (2) Countable Additivity: If $\{A_k\} \subseteq \Sigma$, and $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$P\left(\bigcup_k A_k\right) = \sum_k P(A_k)$$

Proposition:

- (1) If $\{A_k\} \uparrow$ or $\downarrow$ A, then

$$P\left(\lim_{k\to\infty} A_k\right) = \lim_{k\to\infty} P(A_k)$$

- (2) **Inclusion-Exclusion:**

$$P\left(\bigcup_k A_k\right) = \sum_k P(A_k) - \sum_{1\le i<j\le k} P(A_i \cap A_j) + \sum_{1\le i<j<l\le k} P(A_i \cap A_j \cap A_l) - \cdots$$

e.g. Matching

Target: $P(X=0)$

$\Omega = \{\ n_1\ n_2\ \ldots n_N\ \}$ (N! in total)

Let the hats be $\{1,2,\ldots,N\}$, B_k: k-th matched

Then

$$P(X=0) = P\left(\bigcap_k B_k^c\right) = P\left(\left(\bigcup_k B_k\right)^c\right) = 1 - P\left(\bigcup_k B_k\right)$$

$$= 1 - \left(\sum_k P(B_k) - \sum_{i<j} P(B_i \cap B_j) + \cdots\right)$$

$$P(B_k) = \frac{(N-1)!}{N!}, \quad P(B_i \cap B_j) = \frac{(N-2)!}{N!}, \quad \cdots$$

Finally,

$$P(X=0) = 1 - \left[\binom{N}{1}\frac{1}{N} - \binom{N}{2}\frac{1}{N(N-1)} + \cdots\right] = 1 - \left(1 - \frac{1}{2!} + \frac{1}{3!} - \cdots\right)$$

$$\lim_{N\to\infty} P(X=0) = \frac{1}{e}$$

Part 3

P: $\Sigma \to [0,1]$, If Ω is discrete, then $\Sigma = 2^\Omega$

Problem: What if Ω is continuous (e.g., $\mathbb{R}$)?

Consider the interval $(0,1)$. The question is: Can an infinite number of sets be constructed such that the lengths of the sets are equal, and

$$\bigcap_k A_k = \emptyset, \quad \bigcup_k A_k = (0,1)$$

Vitali's Theorem:

- (1) Define "$\sim$" relation.
- (2) Define set S, then

$$(0,1) = S + \{(0,1) \cap \mathbb{Q}\} \mod 1$$

Solution: Simple sets $\xrightarrow{\cap \quad \cup \quad ^c}$ Complex sets

Example: Borel sets: $\{(-\infty, x]\}$

Conclusion: We get **Probability Triple: (Ω, Σ, P)**

Lecture 3

1. Random Variable

Random variable: $\Omega \to \mathbb{R}$, a determined function

$$\Omega \leftrightarrow \Sigma \subseteq 2^{\Omega}, \quad \mathbb{R} \leftrightarrow B \subseteq 2^{\mathbb{R}}$$

$$\mathcal{X} : \Omega \to \mathbb{R}, \quad \forall A \in B, \quad \mathcal{X}^{-1}(A) \in \Sigma$$

$$\mathbf{P}(\boldsymbol{\mathcal{X}} \in \boldsymbol{A}) = \boldsymbol{P}\big(\{\boldsymbol{\omega} \in \boldsymbol{\Omega} \mid \boldsymbol{\mathcal{X}}(\boldsymbol{\omega}) \in \boldsymbol{A}\}\big)$$

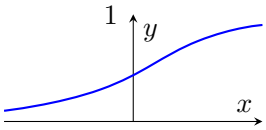
2. Distribution Function & Probability Density

Distribution function:

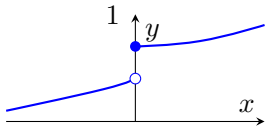
$$\boldsymbol{F_{\mathcal{X}}(x) = P(\mathcal{X} \in x) = P\big(\mathcal{X} \in (-\infty, x]\big)}$$

Characters:

- ① : $F_{\mathcal{X}}(-\infty) = 0, \quad F_{\mathcal{X}}(+\infty) = 1$
- ② : $\forall x \leqslant y, \quad F_{\mathcal{X}}(x) \leqslant F_{\mathcal{X}}(y)$
- ③ : Right Continuous



(a) Continuous



(b) This x has "mass".

Ω — Discrete / Continuous

(1) $\Omega = \{\omega_1, \omega_2, \cdots, \omega_n\}$

$$\mathcal{X}(\Omega) = \{\mathcal{X}(\omega_1), \mathcal{X}(\omega_2), \cdots, \mathcal{X}(\omega_n)\} \xrightarrow{\text{simply written as}} \{x_1, x_2, \cdots, x_n\}$$

$$P(\mathcal{X} \in A) = \sum_{\omega_k \in \mathcal{X}^{-1}(A)} P(\{\omega_k\}) = \sum_{x_k \in A} P(\mathcal{X} = x_k)$$

(2) ω continuum

$$P(\mathcal{X} \in A) = \int_{x \in A} f(x)\, dx$$

$$F_{\mathcal{X}}(x) = P(\mathcal{X} \in (-\infty, x]) = \int_{-\infty}^{x} f_{\mathcal{X}}$$

Define Probability Density:

$$\boldsymbol{f_{\mathcal{X}} = \frac{d}{dx} F_{\mathcal{X}}}$$

Characters:

– ①: $f_{\mathcal{X}}(x) \geqslant 0$

– ②: $\int_{-\infty}^{+\infty} f_{\mathcal{X}} = 1$

Generally, $P(\mathcal{X} = x_k) = 0$

$$P(x < \mathcal{X} < x + \Delta x) = \int_{x}^{x+\Delta x} f_{\mathcal{X}} \approx f_{\mathcal{X}}(x)\Delta x$$

3. Exercises

e.g.1 $\mathcal{X} \sim F_{\mathcal{X}}(x),\ Y = \mathcal{X} + a \quad (a : \text{constant})$

$$F_Y(y) = P(Y \leq y) = P(\mathcal{X} + a \leq y) = P(\mathcal{X} \leq y - a) = F_{\mathcal{X}}(y - a)$$

$$f_Y(y) = f_{\mathcal{X}}(y - a)$$

e.g.2 $\mathcal{X} \sim F_{\mathcal{X}}(x),\ Y = a\mathcal{X} \quad (a \neq 0,\ P(\mathcal{X} = \frac{y}{a}) = 0)$

$$F_Y(y) = P(Y \leq y) = P(a\,\mathcal{X} \leq y) = \begin{cases} P\Big(\mathcal{X} \leq \frac{y}{a}\Big), & a > 0 \\ P\Big(\mathcal{X} \geq \frac{y}{a}\Big), & a < 0 \end{cases}$$

$$F_Y(y) = \begin{cases} F_{\mathcal{X}}\Big(\frac{y}{a}\Big), & a > 0 \\ 1 - F_{\mathcal{X}}\Big(\frac{y}{a}\Big), & a < 0 \end{cases}$$

$$f_Y(y) = \frac{1}{|a|} f_{\mathcal{X}}\Big(\frac{y}{a}\Big)$$

e.g.3 Monte Carlo Simulation

How to generate pseudo-random numbers with a given distribution?

Step1: Generate a uniform distribution

$$U \sim \mathcal{U}(0, 1)$$

$$F_U(t) = \begin{cases} 0, & t < 0 \\ t, & 0 \leq t \leq 1 \\ 1, & t > 1 \end{cases}$$

Step2: Apply the inverse of the given distribution to U.

$$Y = F_{\mathcal{X}}^{-1}(U)$$

$$\begin{aligned} F_Y(y) &= P(Y \leqslant y) = P(F_{\mathcal{X}}^{-1}(U) \leqslant y) \\ &= P(U \leqslant F_{\mathcal{X}}(y)) = F_{\mathcal{X}}(y) \\ &\Rightarrow Y \sim F_{\mathcal{X}} \end{aligned}$$

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L4 Independence

1. (Ω, Σ, P), $A, B \in \Sigma$,

称A.B独立, if $P(A \cap B) = P(A)P(B)$

① if $P(A \cap B) = 0$, 则 A, B 不独立

② if A,B独立, 则 A^c 与 B, A 与 B^c, A^c 与 B^c 独立

2. Binary $\xrightarrow{\text{扩展}}$ Multiple

对 $A_1, \cdots, A_n \in \Sigma$, 称 $A_1, \cdots, A_n$ 互相独立 (Mutual Independent),

if $P(\bigcap_{k=1}^{n} A_k) = \prod_{k=1}^{n} P(A_k)$

注: 两两独立 $\not\Leftrightarrow$ 互相独立

e.g. (#) Coin Tossing n times $X_1, \cdots, X_n$

构造 $A_1 = \{X_1 = X_2\}$, $A_2 = \{X_2 = X_3\}$, $\cdots$, $A_n = \{X_n = X_1\}$

$P(A_i \cap A_j) = \frac{1}{4} = P(A_i)P(A_j)$ 两两独立

$P(\bigcap_{k=1}^{n} A_k) = \frac{1}{2^{n-1}} \neq \frac{1}{2^n} = \prod_{k=1}^{n} P(A_k)$, 不互相独立

3. Events $\xrightarrow{\text{扩展}}$ Random Variables

称 $X_1, \cdots, X_n$ 独立, if $P(X_1 \le x_1 \odot X_2 \le x_2 \odot \cdots \odot X_n \le x_n) = \prod_{k=1}^{n} P(X_k \le x_k)$

(逗号表示 $\cap$, 且)

注: 上式很强大, ∵ 可对 x_i 取 ∞, 则左侧相当于其余 $\cap$ 全集, 右侧相当于其余乘1.

从而可得到任意若干个 X 独立

4. e.g. | AI $\{w_1, \cdots, w_n\}$ $X: \{w_1, \cdots, w_n\} \to \{0,1\}$

A: Standard Answer 定义 $L_n(X) = \{w \in \Omega : X(w) \neq A(w)\}$ 显然, $L_n(X) \neq \phi$

希望 $P(L_n(X)) \le \varepsilon$ 且要保证 对于 $w_k \in \{w_1, \cdots, w_n\}$, $X(w_k) = A(w_k)$

但是对于没调好的神经网络, 有可能 $P(L_n(X)) > \varepsilon$

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此时我们设：$P\left(P(L_n(x_1)>\varepsilon)\right)=\delta$，问：n多大时，$\delta$足够小？

$$P\left(P(L_n(x_1)>\varepsilon)\right) \le P\left(\bigcap_{k=1}^{n} X(w_k)=A(w_k),\ k=1,2,\cdots,n\right)$$

$$= P\left(\bigcup_{m=1}^{N}\bigcap_{k=1}^{n} X_m(w_k)=A(w_k),\ k=1,2,\cdots,n\right) \le \sum_{m=1}^{N} P\left(\bigcap_{k=1}^{n} X_m(w_k)=A(w_k),\ k=1,2,\cdots,n\right)$$

Independent $\sum_{m=1}^{N}\prod_{k=1}^{n} P\left(X_m(w_k)=A(w_k)\right) \le N(1-\varepsilon)^n \le N e^{-\varepsilon n}$

$\le 1-\varepsilon$

$\Rightarrow \delta \le N e^{-\varepsilon n}$ $\log\delta + \log N \le -\varepsilon n$

可见：$\delta\downarrow$，$\varepsilon\downarrow$ 则需要 $n\uparrow$

$N\uparrow$ $n\uparrow$

5、定义 Indicator：$A\subset\Sigma$，$I_A(w)=\begin{cases}1 & w\in A\\ 0 & w\notin A\end{cases}$

应用1：定义 Δ：$A\Delta B=(A\cap B^c)\cup(B\cap A^c)$



求证：$(A\Delta B)\Delta C = A\Delta(B\Delta C)$

证：$I_{A\Delta B}(w)=I_A(w)\oplus I_B(w)$ 而 $\oplus$ 是 s.t. 结合律的

应用2（Independence e.g.2：Borel Cantelli）

① 对 $A_1, A_2, \cdots, A_n, \cdots$ 证：$\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k \supseteq \bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k$

证：$I_{\bigcap_k A_k}(w)=\min_k I_{A_k}(w)$，$I_{\bigcup_k A_k}(w)=\max_k I_{A_k}(w)$

$\Rightarrow I_{\bigcap_n\bigcup_k A_k}(w)=\min_n\max_{k\ge n} I_{A_k}(w)$ / limsup $\Rightarrow I_{\cap\cup} \ge I_{\cup\cap} \Rightarrow \cap\cup A_k \supseteq \cup\cap A_k$

$I_{\bigcup_n\bigcap_k A_k}(w)=\max_n\min_{k\ge n} I_{A_k}(w)$ / liminf

② 命题：(i) 设 $\sum_k P(A_k)<\infty$，则 $P\left(\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k\right)=0$

证：令 $B_n=\bigcup_{k=n}^{\infty} A_k$，则 $B_n\downarrow$

$$\Rightarrow P\left(\bigcap_{n=1}^{\infty} B_k\right)=\lim_{n\to\infty} P(B_n) \le \lim_{n\to\infty}\left(\sum_{k=n}^{\infty} P(A_k)\right)=0$$

(ii) 设 $\sum_k P(A_k)=\infty$ 且 independent，则 $P\left(\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k\right)=1$

$\Leftarrow P\left(\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k^c\right)=0$ $\because P\left(\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k^c\right) \le \sum_{n=1}^{\infty} P\left(\bigcap_{k=n}^{\infty} A_k^c\right)$ $\therefore \Leftarrow P\left(\bigcap_{k=n}^{\infty} A_k^c\right)=0,\ \forall n$

$P\left(\bigcap_{k=n}^{\infty} A_k^c\right) \overset{\text{Independent}}{=\!=} \prod_{k=n}^{\infty} P(A_k^c)=\prod_{k=n}^{\infty}(1-P(A_k)) \le \prod_{k=n}^{\infty}\exp(-PA_k)=\exp\left(-\sum_{k=n}^{\infty}(PA_k)\right)=0$ $\therefore$ ✓

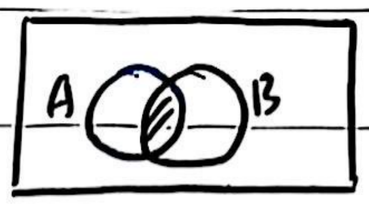
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L5 Conditional Probability

(Ω, Σ, P) $A, B \in \Sigma$ $P(B|A) = \dfrac{P(A \cap B)}{P(A)}$

Another sample space



- Independence: $P(A \cap B) = P(A)P(B) \Rightarrow P(B|A) = P(B)$

e.g. X, Y, Z r.v. i.i.d (独立同分布)

Q: $P(X>Y) = P(Y>X) = \frac{1}{2}$, $P(X>Y>Z) = \cdots = \frac{1}{6}$

算 $P(X>Z|X>Y) = \dfrac{P(X>Z, X>Y)}{P(X>Y)} = \dfrac{1/3}{1/2} = \frac{2}{3} > \frac{1}{2} = P(X>Z)$

① Multiplication Formula

$$P(A_1 A_2 \cdots A_n) = P(A_n | A_{n-1} \cdots A_1) P(A_{n-1} \cdots A_1)$$
$$= P(A_n | A_{n-1} \cdots A_1) P(A_{n-1} | A_{n-2} \cdots A_1) P(A_{n-2} \cdots A_1) = \cdots$$
$$= \left(\prod_{k=2}^{n} P(A_k | A_{k-1} \cdots A_1) \right) P(A_1)$$

e.g. Matching 求: $P(X=k)$

$$P(X=k) = \binom{n}{k} P(A_1 A_2 \cdots A_k \tilde{A}_{k+1} \cdots \tilde{A}_n)$$
$$= \binom{n}{k} P(A_1) P(A_2|A_1) P(A_3|A_1A_2) \cdots P(A_k|A_1 \cdots A_{k-1}) P(\tilde{A}_{k+1} \cdots \tilde{A}_n | A_1 \cdots A_k)$$
$$= \frac{n!}{(n-k)!\,k!} \cdot \frac{1}{n} \cdot \frac{1}{n-1} \cdot \frac{1}{n-2} \cdots \frac{1}{n-k+1} P_{n-k}(X=0)$$
$$= \frac{1}{k!} P_{n-k}(X=0)$$

② Addition Formula (Total Probability)

$A, B_1, \cdots, B_n \in \Sigma$, $B_1 \cup \cdots \cup B_n = \Omega$ $B_i \cap B_j = \phi$ $(\forall i \neq j)$

↓ "不漏" ↓ "不重"

见: $P(A) = \sum_{k=1}^{n} P(AB_k) = \sum_{k=1}^{n} P(A|B_k) P(B_k)$ ("分情况讨论")

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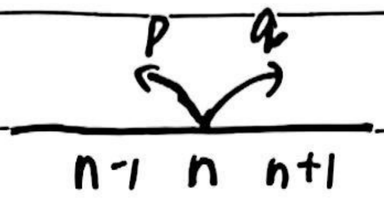
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e.g. Urn m white n black (/买彩票)

$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$

$= \frac{m-1}{m+n-1}\frac{m}{m+n} + \frac{m}{m+n-1}\frac{n}{m+n} = \frac{m}{m+n}$

e.g. Gambler Ruin N dollars



问：输光的概率

方法：Recursive (递推)

"First Step" (先走一步再看)

$P_n = pP_{n-1} + qP_{n+1}$ (注：之所以能写成这种形式，是因为该问题有马尔科夫性)

$z = pz^2 + q \quad z_1 = 1, z_2 = \frac{q}{p}$

(1) $z_1 \neq z_2 \quad P_n = a_1 \cdot 1^n + a_2 (\frac{q}{p})^n = \begin{cases} q<p: \text{解出 } a_1, a_2 \text{ 即可} \\ q>p: a_2 = 0, a_1 = 1 \text{ 必输光} \end{cases}$

(2) $z_1 = z_2$ 即 $p/q = 1 \quad P_n = (a_1 + a_2 n) \cdot 1^n$

☆ Bayesian Formular $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$ prior 先验 → likelihood 似然 → posterior 后验

$P(B) \quad P(A|B) \quad P(B|A)$

$= \frac{P(A|B)P(B)}{\sum_k P(A|C_k)P(C_k)}$

e.g. Urn1 m white n black Urn2 n white m black (m → n)

$P(w_2|w_1) = P(w_2U_1|w_1) + P(w_2U_2|w_1)$ Hint: $P(C|BA) = P(C|BA)P(B|A)$

$= P(w_2|U_1w_1)P(U_1|w_1) + P(w_2|U_2w_1)P(U_2|w_1)$

$= P(w_2|U_1)P(U_1|w_1) + P(w_2|U_2) \cdot P(U_2|w_1)$

$= \frac{m}{m+n}P(U_1|w_1) + \frac{n}{m+n}P(U_2|w_1)$

贝叶斯公式：完成了"因果转换"

$P(U_1|w_1) = \frac{P(w_1|U_1)P(U_1)}{P(w_1)} = \frac{P(w_1|U_1)P(U_1)}{P(w_1|U_1)P(U_1) + P(w_1|U_2)P(U_2)} = \frac{\frac{m}{m+n}\cdot\frac{1}{2}}{\frac{m}{m+n}\cdot\frac{1}{2} + \frac{n}{m+n}\cdot\frac{1}{2}} = \frac{m}{m+n}$

$\Rightarrow P(w_2|w_1) = \frac{m^2+n^2}{(m+n)^2} > \frac{1}{2} = P(w_1)$

No.

Date

L6 Discrete Random Variables

1. Bernoulli (Two-Point)

$\Omega = \{w_1, w_2\}$ $X \in \{0,1\}$ $P(X=1)=p$, $P(X=0)=q$, $p,q \geq 0$ 且 $p+q=1$

$\Rightarrow E(X) = \sum_k x_k P(X=x_k) = 1\cdot p + 0\cdot q = p$

if $p=q=\frac{1}{2}$, $E(X)=\frac{1}{2}$

2. Binomial

$\Omega = \{w_1, w_2, \cdots, w_{n+1}\}$ Finite, $X(w_k)=x_k$, $x_k \in \{0,1,\cdots,n\}$

Shooting Model

$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$ (易验证: $\sum_k P(X=k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p+1-p)^n = 1$)

$E(X) = \sum_{k=0}^{n} x_k P(X=x_k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$

$k\binom{n}{k} = n\binom{n-1}{k-1}$

$\sum_{k=1}^{n} n \binom{n-1}{k-1} p^k (1-p)^{n-k} = \sum_{k=1}^{n} np \binom{n-1}{k-1} p^{k-1} (1-p)^{n-1-(k-1)} = \underline{np}$ (Linear)

if $p=\frac{1}{2}$, $E(X)=\frac{n}{2}$ 问: $n\to\infty$, $P(X=\frac{n}{2})$ ↑ ?

设 $n=2m$, $P(X=\frac{n}{2}) = P(X=m) = \binom{2m}{m}\left(\frac{1}{2}\right)^{2m} = \frac{(2m)!}{m!m!}\left(\frac{1}{2}\right)^{2m}$

$n! \sim \left(\frac{n}{e}\right)^n \sqrt{2\pi n}$ $\frac{(2m/e)^{2m}\cdot\sqrt{4\pi m}}{(m/e)^{2m}\cdot 2\pi m}\left(\frac{1}{2}\right)^{2m} \sim \frac{1}{\sqrt{m}}$ ∴ ✗

3. Poisson

令 Binomial 中: $n\to\infty$, $p\to 0$, $np=\lambda$, 考虑 $P(X=k)$

$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n(n-1)\cdots(n-k+1)}{k!}\left(\frac{\lambda}{n}\right)^k \left(1-\frac{\lambda}{n}\right)^{n-k}$

$= \frac{\lambda^k}{k!} \frac{n(n-1)\cdots(n-k+1)}{n^k} \left(1-\frac{\lambda}{n}\right)^{-k} \left(1-\frac{\lambda}{n}\right)^{n} \xrightarrow{n\to\infty} \underline{\frac{\lambda^k}{k!} e^{-\lambda}}$ Poisson

易验证 $E(X) = \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda} = \lambda$ ($\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$)

Note: $P(X \text{ is even}) = \sum_{k \text{ even}} \frac{\lambda^k}{k!} e^{-\lambda} = \frac{e^{\lambda}+e^{-\lambda}}{2} e^{-\lambda} = \frac{1}{2} + \frac{1}{2} e^{-2\lambda}$

$P(X \text{ is even}) > P(X \text{ is odd})$

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No.

Date

4. Geometrical Memoryless

Shooting Model: $P(X=k) = p \cdot q^{k-1}$

$$E(X) = \sum_{k=1}^{\infty} k p q^{k-1} = p\left(\sum_{k=1}^{\infty} q^k\right)' = \frac{1}{p}$$

e.g. 3 Dice: 2 Normal, 1 Fault (the same number)

记 A_k: A在k次中发生变化, B_k: B …

① P(两次找出Fault) $= P(A_2B_2) = P(A_2)P(B_2) = \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$

② P(三次 | 两次) $= P(A_3B_3 | (A_2B_2)^c) = \frac{P(A_3B_3 | (A_2^c \cup B_2^c))}{P((A_2B_2)^c)} = \frac{P(A_3B_3A_2^c \cup A_3B_3B_2^c)}{1 - P(A_2B_2)}$

$$= \frac{P(A_3B_3A_2^c) + P(A_3B_3B_2^c) - P(A_3B_3A_2^cB_2^c)}{1 - P(A_2B_2)}$$

$P(A_3B_3A_2^c) = P(A_3A_2^c)P(B_3) = \frac{1}{6} \times \frac{5}{6} \times \left(1 - \frac{1}{36}\right) = \frac{35 \times 5}{36^2}$

原 $= \frac{\frac{35 \times 5 \times 2}{36^2} - \frac{25}{36^2}}{1 - \frac{25}{36}} = \frac{325}{396}$

③ E(找出Fault需要的次数) $= 1 + \frac{6}{5} + \frac{6}{5} = \frac{17}{5}$

(由 Geometrical $E(X) = \frac{1}{p}$, $E_{几何}$(one dice) $= \frac{1}{\frac{5}{6}} = \frac{6}{5}$)

思考一: 设 $E = 1 + E'$

则 $E' = \frac{25}{36} \times 1 + \frac{10}{36} \times (E_{几何}(\text{one dice}) + 1) + \frac{1}{36}(E' + 1)$

$\Rightarrow E' = \frac{48}{35}$ $E = 1 + \frac{48}{35}$

思考二: 记 $P_{k,q}$ 表示除去第一次, 扔k次后有q个骰子的状态与扔最开始那次相比改变过的概率, 其中 $q \in \{0,1,2\}$, $k \in \mathbb{Z}^+$ 且截止到q恰=2 (即 $P_{k,2} = P_{min[k],2}$)

则 $E = 1 + k \cdot \left(\sum_{k=1}^{\infty} P_{k,2}\right)$

$P_{k,2} = \frac{5}{6} P_{k-1,1} + \frac{25}{36} P_{k-1,0}$ 其中 $P_{k-1,0} = \left(\frac{1}{36}\right)^{k-1}$

$P_{k-1,1} = \frac{1}{6} P_{k-2,1} + \frac{10}{36} P_{k-2,0}$

No.

Date

$P_{k-1,1} = \frac{1}{6} P_{k-2,1} + \frac{10}{36} \left(\frac{1}{36}\right)^{k-2}$

$\Rightarrow P_{k-1,1} = 2\left(\frac{1}{6}\right)^{k-1} - \frac{1}{18}\left(\frac{1}{36}\right)^{k-2}$

$\Rightarrow P_{k,2} = \frac{5}{6}\left[2\left(\frac{1}{6}\right)^{k-1} - \frac{1}{18}\left(\frac{1}{36}\right)^{k-2}\right] + \frac{25}{36}\left(\frac{1}{36}\right)^{k-1}$

$= 10\left(\frac{1}{6}\right)^{k} - 35\left(\frac{1}{36}\right)^{k}$

$E = 1 + \sum_{k=1}^{\infty} k \cdot \left[10\left(\frac{1}{6}\right)^{k} - 35\left(\frac{1}{36}\right)^{k}\right]$

$= 1 + \frac{48}{35}$

思考三：

$P_n = P(\max(X_A, X_B) = n) = P(\max(X_A, X_B) \le n) - P(\max(X_A, X_B) \le n-1)$

$P(\max(X_A, X_B) \le n) = P(X_A \le n, X_B \le n) = P(X_A \le n)\, P(X_B \le n) = \left(1 - \left(\frac{1}{6}\right)^{n}\right)^2$

$\Rightarrow P_n = \left(1 - \left(\frac{1}{6}\right)^{n}\right)^2 - \left(1 - \left(\frac{1}{6}\right)^{n-1}\right)^2$

$E = 1 + \sum_{k=1}^{\infty} k P_k = 1 + \frac{48}{35}$

另：$P(\max(X_A, X_B) = n)$ ⟨ ① $\frac{5}{6}\left(\frac{1}{6}\right)^{n-1} \cdot \sum_{i=1}^{n} \frac{5}{6}\left(\frac{1}{6}\right)^{i-1}$; ② $\sum_{i=1}^{n-1} \frac{5}{6}\left(\frac{1}{6}\right)^{i-1} \cdot \frac{5}{6}\left(\frac{1}{6}\right)^{n-1}$

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L7 Continous Distribution

Randomness ⇔ Entropy

"Typical Distribution" ⇔ Most "Random" ⇔ Maximum Entropy

定义 $H(f_x) = \int_I f_x \log_2 f_x$

I { $[a,b]$ $f_x \geq 0$, $\int_{-\infty}^{\infty} f_x = 1$

$[0,\infty)$ $f_x \geq 0$, $\int_{-\infty}^{\infty} f_x = 1$, $\int_{-\infty}^{\infty} x f_x(x) dx = A$

$(-\infty,\infty)$ .., .., .., $\int_{-\infty}^{\infty} x^2 f_x(x) dx = B$

考虑 $\max_{f_x} H(f_x)$

① $I=(-\infty,\infty)$ f_x s.t. $f_x \geq 0$, $\int_{-\infty}^{\infty} f_x = 1$, $\int_{-\infty}^{\infty} x f_x = A$, $\int_{-\infty}^{\infty} x^2 f_x = B$

Lemma: 设 $f, g \geq 0$, $\int_{-\infty}^{\infty} f = \int_{-\infty}^{\infty} g = 1$, 则 $\int_{-\infty}^{\infty} f \log \frac{f}{g} \geq 0$ (Kullback-Leibler Divergence)

证: $\int_{-\infty}^{\infty} f \log \frac{f}{g} \geq \int_{-\infty}^{\infty} f \left(1 - \frac{1}{f/g}\right) = \int_{-\infty}^{\infty} (f - g) = 0$ Q.E.D

$\Rightarrow -\int_{-\infty}^{\infty} f_x \log \frac{f_x}{g} \leq 0$

$\Rightarrow -\int_{-\infty}^{\infty} f_x \log f_x \leq -\int_{-\infty}^{\infty} f_x \log g$ 取 $g = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-u)^2}{2\sigma^2}\right)$ $\log \sqrt{2\pi}\sigma + \int_{-\infty}^{\infty} f_x \frac{(x-u)^2}{2\sigma^2}$

找 g s.t. 这项 = const

取 $u = A$ (均值) $\log \sqrt{2\pi}\sigma + \frac{1}{2\sigma^2} \int_{-\infty}^{\infty} f_x (x-A)^2 = \log \sqrt{2\pi}\sigma + \frac{1}{2\sigma^2} Var(X)$

由 $Var(X) = EX^2 - (EX)^2 = B - A^2$ $\therefore$ 取 $\sigma^2 = B - A^2$

$\Rightarrow -\int_{-\infty}^{\infty} f_x \log f_x \leq \log \sqrt{2\pi e}\sigma$ $\therefore$ Gaussian

② $I=[0,\infty)$

$-\int_0^{\infty} f_x \log f_x \leq -\int_0^{\infty} f_x \log g$ 取 $g = \lambda \exp(-\lambda x)$ $\int_0^{\infty} f_x (\log\lambda - \lambda x) = \log\lambda - \lambda A$

取 $\lambda = \frac{1}{A}$ $\therefore$ Exponential

③ $I=[a,b]$

$-\int_a^b f_x \log f_x \leq -\int_a^b f_x \log g$ 取 $g = \frac{1}{b-a}$ $\log \frac{1}{b-a}$ $\therefore$ Uniform

1. Uniform $[a,b]$ $f_x = \begin{cases} \frac{1}{b-a} & x \in [a,b] \\ 0 & \text{else} \end{cases}$

e.g. $X \sim U(0,1) \Rightarrow 1-X \sim U(0,1)$

No.

Date

思路：$X \sim f_X$，$Y = h(X)$（h可逆），则 $F_Y(y) = P(Y \le y) = P(h(X) \le y)$ 不动方h↑

（known）（unknown） $P(X \le h^{-1}(y)) = F_X(h^{-1}(y))$ 注：$(h^{-1})'|_y = \frac{1}{h'|_{h^{-1}(y)}}$

$\Rightarrow F_Y(y) = P(Y \le y) = P(1-X \le y) = P(X \ge 1-y) = 1 - P(X < 1-y) = 1 - F_X(1-y)$

$= 1 - (1-y) = y$ 可见，$Y = 1-X$ 仍是均匀分布

e.g.2 n independent X_k $(k = 1, \cdots, n)$

$P(L > x) = P(X_1 > x, \cdots, X_n > x) = \prod_{k=1}^{n} P(X_k > x) = \prod_{k=1}^{n} (1 - F_{X_k}(x)) = (1-x)^n$

$\|$

$1 - P(L \le x) = 1 - F_L(x)$

$\Rightarrow F_L(x) = 1 - (1-x)^n$ $n \ge 2$ 不是均匀分布

2. Exponential $X \sim E(\lambda)$ $f_X = \begin{cases} \lambda e^{-\lambda x} & x \ge 0 \\ 0 & x < 0 \end{cases}$

- $E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx = \frac{1}{\lambda}$
- Memoryless：$P(X > x+y \mid X > x) = P(X > y)$

证：$P(X > x+y \mid X > x) = \frac{P(X > x+y,\ X > x)}{P(X > x)} = \frac{P(X > x+y)}{P(X > x)}$

由 $F_X(x) = 1 - e^{-\lambda x} = P(X \le x) \Rightarrow P(X > x) = e^{-\lambda x}$

$\therefore P(X > x+y \mid X > x) = \frac{e^{-\lambda(x+y)}}{e^{-\lambda x}} = e^{-\lambda y} = P(X > y)$

注：也可∈，只须证：$H(x+y) = H(x)H(y) \Rightarrow H(x) = Ae^{-\lambda x}$（设 $H \ne 0$）

（证：$H(x+h) = H(x)H(h) \Rightarrow \lim_{h \to 0} \frac{H(x+h) - H(x)}{h} = H'(x) = \lim_{h \to 0} \frac{H(h) - 1}{H(h)} H(x)$

$\because H(0) = H(0)$ 且 $H \ne 0$ $\therefore H(0) = 1 \Rightarrow RHS = \lambda H(x) \Rightarrow H(x) = Ae^{-\lambda x}$）

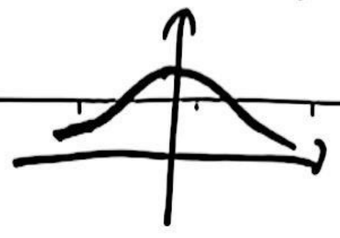
e.g. 设 X, Y i.i.d $Exp(\lambda)$，问：$\min(X, Y) \sim ?$

$P(\min(X, Y) \le t) = 1 - P(\min(X, Y) > t) = 1 - P(X > t, Y > t) = 1 - P(X > t)P(Y > t)$

$= 1 - e^{-2\lambda t}$

$\Rightarrow \min(X, Y) \sim Exp(2\lambda)$

3. Gaussian $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{(x-u)^2}{2\sigma^2}\right)$

$N(u, \sigma^2)$ $N(0, 1)$  ∴ $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$

Date

L8 Expectation

X { 离 Discrete $E(X) = \sum_k x_k P(X = x_k)$

离 Continous $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$

1. 性质：① Linear $E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n)$ (No condition)

e.g.1 $X \sim B(n,p)$, 用 Linear 算 $E(X) = np$

X 拆分 $X_1 + \cdots + X_n$ $X_k = \begin{cases} 1 & : p \\ 0 & : 1-p \end{cases}$ $\Rightarrow E(X) = E(\sum_{k=1}^{n} X_k) = \sum_{k=1}^{n} E(X_k)$

$= np$

e.g.2 Matching X: number of Matched

$X = X_1 + \cdots + X_n$ $X_k = \begin{cases} 1 & k\text{th matched} \\ 0 & \text{else} \end{cases}$ $: p = \frac{(n-1)!}{n!} = \frac{1}{n}$

$E(X) = E(\sum_{k=1}^{n} X_k) = \sum_{k=1}^{n} E(X_k) = n \cdot \frac{1}{n} = 1$

e.g.3 Runs (游程) $\underbrace{|00|0011|01|\cdots}_{n}$ 1: p 0: $1-p$ independent N: number of Runs

Key Point: 不是每一位是什么，而是"缝隙"，即是否改变

$\Rightarrow E(N) = E(\sum_{k=1}^{n-1} Y_k) + 1$ $Y_k = \begin{cases} 1 & : 2p(1-p) \\ 0 \end{cases}$

$= 2(n-1)p(1-p) + 1$

e.g.4 Couples n对couples, m persons out, 问：还剩多少对couples

运用 Linear 的关键：把复杂对象拆成简单对象的叠加，而不必关心这些对象之间的关系

$X = X_1 + \cdots + X_n$ X_k { 1 这对还在 $: \binom{2n-2}{m} / \binom{2n}{m}$

0 else

$\Rightarrow E(X) = n \cdot \binom{2n-2}{m} / \binom{2n}{m}$

e.g.5 Simple Diffusion R: n red H: n black (设为 X)

每次从两个篮子各拿一个球，交换，再放回。操作 m 次，问 R 中有几个红球

Key Point: 对一个进行追踪 $X = X_1 + \cdots + X_n$ $X_k = \begin{cases} 1 & \text{kth red stay in R} \\ 0 & \text{else} \end{cases}$

$P(\text{一个球被选中 } i \text{ 次}) = \binom{m}{i} (\frac{1}{n})^i (1-\frac{1}{n})^{m-i}$

$\Rightarrow P(X_k = 1) = \sum_{i:\text{even}}^{m} \binom{m}{i} (\frac{1}{n})^i (1-\frac{1}{n})^{m-i} = \frac{1}{2}\left[(\frac{1}{n} + 1 - \frac{1}{n})^m + (-\frac{1}{n} + 1 - \frac{1}{n})^m\right]$ ※

① kth 被选中了偶数次

$\Rightarrow E(X) = \frac{n}{2}\left[1 + (1 - \frac{2}{n})^m\right]$

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② $X \sim f_X(x)$ $Y = g(X)$ $E(Y) = ?$

$F_Y(y) = P(Y \le y) = P(g(X) \le y) \overset{?}{=} P(X \overset{?}{\le} g^{-1}(y))$ 难！

⇒直接算 $E(Y) = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$ (No condition on g)

e.g.1 $\theta \sim U(0, 2\pi)$ $X = \cos(\theta)$, $E(X) = \int_0^{2\pi} \frac{1}{2\pi} \cos\theta\, d\theta = 0$

e.g.2 $X \sim U(0,1)$ $Y = \min\{X, 1-X\}$ $E(Y) = \int_0^1 \min\{x, 1-x\}dx = \int_0^{\frac{1}{2}} x\,dx + \int_{\frac{1}{2}}^1 (1-x)dx = \frac{1}{4}$

e.g.3 $X \sim U(0,1)$ $Y = a^X$ $E(Y) = \int_0^1 a^x dx = \frac{a-1}{\ln a}$

$$F_Y(y) = P(Y \le y) = P(a^X \le y) = \begin{cases} P(X \le \log_a y) = F_X\left(\frac{\ln y}{\ln a}\right) & a > 1 \\ P(1 \le y) & a = 1 \\ P(X \ge \log_a y) = 1 - F_X\left(\frac{\ln y}{\ln a}\right) & 0 < a < 1 \end{cases}$$

注：①. ② 从 $E(X) = \int_\Omega X(\omega) P(d\omega)$ 的角度看是显然的

2. 设 $g(x)$ convex (定义 convex: $g(x) \ge g(x_0) + L_{x_0}(x - x_0)$ ∪), 则

$E(g(X)) \ge g(EX)$ (Jensen) $g(EX)$ 0

证：Let $x_0 = E(X) \Rightarrow g(x) \ge g(E(X)) + L(x - EX) \Rightarrow E(g(X)) \ge E(g(E(X))) + E(L(X - EX))$

3. 定义 $Var(X) = EX^2 - (EX)^2$ 理解：① Jensen 的 gap

$= E(X - EX)^2$ ② 用一个确定数对随机变量做逼近

定义 Metric on Variables: $d(X,Y) = (E(X-Y)^2)^{\frac{1}{2}}$ (Mean Square Distance)

考虑 $\min_a d(X,a)$: (⇒ $\min_a (E(X-a)^2)^{\frac{1}{2}}$ ⇒ $\min_a E(X-a)^2$ ⇒ $\frac{d}{da} E(X-a)^2 = 0$

⇒ $a = EX$

从而定义 $Var(X) = E(X - EX)^2$

4. 命题1：设 X, Y independent, 则 $E(XY) = EXEY$, 则 $E(X - EX)(Y - EY) = 0$

命题2：设 $X_1, \cdots, X_n$ i.i.d, $\bar{X} = \frac{1}{n}\sum_{k=1}^n X_k$,

则 $Var(\bar{X}) = E\left(\frac{1}{n}\sum X_k - \frac{1}{n}\sum E(X_k)\right)^2 = \frac{1}{n^2} E\left(\sum X_k - \sum EX_k\right)^2 = \frac{1}{n^2} E\left(\sum (X_k - EX_k)\right)^2$

$= \frac{1}{n^2} E\left(\sum (X_k - EX_k)^2 + \sum_{i \ne j} (X_i - EX_i)(X_j - EX_j)\right)$

$\because E(X_i - EX_i)(X_j - EX_j) = E(X_i - EX_i) E(X_j - EX_i) = 0$

$\therefore Var(\bar{X}) = \frac{1}{n^2} E\left(\sum (X_k - EX_k)^2\right) = \frac{1}{n^2} \sum E(X_k - EX_k)^2 = \frac{1}{n} E(X_1 - EX_1)^2$

Campus (⇒ 多次采样取平均值有利于减小方差)

L9 More on Expectation

1. 回顾，Magic 1： $E(X_1+\cdots+X_n)=E(X_1)+\cdots+E(X_n)$

e.g. 每条边存在的概率都为 p，且 independent，

问：① 组成三角形个数的期望 ② 方差

解：① 设个数为 N，最多 $\binom{4}{3}=4$ 个，$N=I_1+I_2+I_3+I_4$ $I_k=\begin{cases}1 & \exists\ p^3\\ 0 & \exists\ 1-p^3\end{cases}$

$\Rightarrow E(N)=4p^3$

② $Var(N)=E(N^2)-(EN)^2$

$E(N^2)=E(I_1+I_2+I_3+I_4)^2=\sum_{k=1}^{4}E(I_k^2)+\sum_{i\neq j}^{12}E(I_iI_j)$

$I_k^2=\begin{cases}1 & p^3\\ 0 & 1-p^3\end{cases}$ $I_iI_j\ (i\neq j)=\begin{cases}1 & p^5\\ 0 & 1-p^5\end{cases}$ （∵ 4两个0，都共且仅共一条边）

$\Rightarrow E(N^2)=4p^3+12p^5$ $\Rightarrow Var(N)=4p^3+12p^5-16p^6$

Magic 2： $X\sim f_X(x)$，$Y=g(X)$，则 $E(Y)=\int_{-\infty}^{\infty}gf_X$

e.g. $X,Y\overset{i.i.d}{\sim}N(0,1)$ $M=\max(X,Y)$，$L=\min(X,Y)$，问：$E(M)$，$E(L)$

解：$\max(X,Y)=\frac{X+Y+|X-Y|}{2}$，$\min(X,Y)=\frac{X+Y-|X-Y|}{2}$

$\therefore\begin{cases}M+L=X+Y & E(X+Y)=0\\ M-L=|X-Y| & E(|X-Y|)=?\end{cases}$ Lemma： $X-Y\sim N(0,2)$

$\therefore E|X-Y|=\int_{-\infty}^{\infty}|x|\frac{1}{\sqrt{2\pi}\sqrt{2}}\exp(-\frac{x^2}{4})dx=\frac{2}{2\sqrt{\pi}}\int_0^{\infty}xe^{-\frac{x^2}{4}}dx=\frac{2}{\sqrt{\pi}}$

$\Rightarrow\begin{cases}E(M)=\frac{1}{\sqrt{\pi}}\\ E(L)=-\frac{1}{\sqrt{\pi}}\end{cases}$ 记！

2. $E(\frac{X-\mu}{\sigma})^2=1$ ⟶ 方差

(矩) $E(\frac{X-\mu}{\sigma})^3=0$ ⟶ 偏度 (skewness)

命题1：X is symmetric $\Leftrightarrow f(x)=f(2\mu-x)\Leftrightarrow f(\mu+x)=f(\mu-x)$

命题2：$X\overset{d}{=}-X\Rightarrow f(x)=f(-x)\Rightarrow E(X^{2k+1})=\int_{-\infty}^{\infty}x^{2k+1}f(x)dx=0$

$E(\frac{X-\mu}{\sigma})^4$ ⟶ 峰度 (kurtosis)

3. 中位数 (Median)

定义：m is Median $\Leftrightarrow P(X \geq m) = \frac{1}{2}$, $P(X \leq m) = \frac{1}{2}$

$$\begin{cases} E(X) = \arg\min_c E|X-c|^2 \\ \text{med}(X) = \arg\min_c E|X-c| \end{cases} \quad (\star)$$

证($\star$)：只须证：$E|X-c| - E|X-m| \geq 0$, $\forall c$

不妨设 $m \leq c$, 则 $\begin{cases} X \leq m \text{ 时}, & |X-c| - |X-m| = c-X-(m-X) = c-m \geq 0 \\ X > m \text{ 时}, & |X-c| - |X-m| \geq X-c-(X-m) = m-c \end{cases}$

即 $Y = |X-c| - |X-m| \geq \begin{cases} c-m, & X \leq m \\ m-c, & X > m \end{cases}$

$$\Rightarrow E(Y) = E(Y I_{\{X \leq m\}}) + E(Y(1-I_{\{X \leq m\}})) \geq (c-m)E(I_{\{X \leq m\}}) + (m-c)E(1-I_{\{X \leq m\}})$$

$$= (c-m)P(X \leq m) + (m-c)(1-P(X \leq m)) = \underbrace{(c-m)}_{\geq 0}\underbrace{(2P(X \leq m)-1)}_{\geq 0} \geq 0$$

4. Moment Generating Functions (矩母函数)

$$G_X(t) = E(\exp(tX)) \begin{cases} \sum_k e^{tx_k} P(X = x_k) \\ \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx \end{cases}$$

- $X \sim \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix} \Rightarrow G(t) = pe^t + 1-p$
- $X \sim B(n,p) \Rightarrow G(t) = \sum_{k=0}^{n} e^{tk} p^k (1-p)^{n-k} \binom{n}{k} = (pe^t + 1-p)^n$
- $X \sim N(0,1) \Rightarrow G(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{x^2}{2}}\,dx = e^{\frac{t^2}{2}}$

又观察 $G_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx$, 这 $\simeq$ Laplace $\therefore f_X$ 与 G_X 一一对应

$t \to jw$, 定义 $\phi_X(w) = E(e^{jwx})$ 为 Characteristic Function

e.g. $X, Y \overset{i.i.d}{\sim} N(0,1) \Rightarrow G_{X+Y}(t) = E(e^{t(X+Y)}) = E(e^{tX} \cdot e^{tY})$

$$\overset{\text{independent}}{=\!=} E(e^{tX}) \cdot E(e^{tY}) = G_X(t) \cdot G_Y(t) = e^{t^2} \quad (X+Y \sim N(0,2))$$

(变换域相乘 $\leftrightarrow$ 原始域相卷)

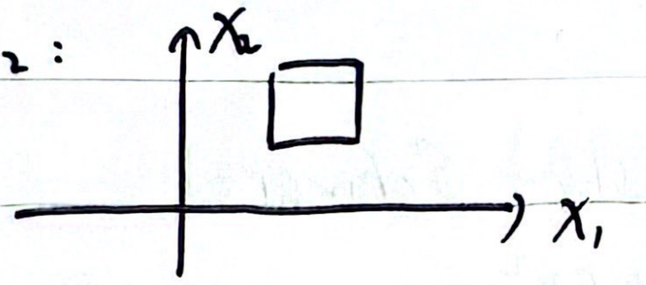
key point: "以貌取人" ($\because$ 一一对应)

L10 Multiple Random Variables

一元 X : $F_X(x) = P(X \le x)$, $f_X(x) = \frac{d}{dx} F_X(x)$

$P(X \in [a,b]) = F_X(b) - F_X(a)$ $\quad P(X \in A) = \int_A f_X(x)\,dx$

二元 X_1, X_2:



(PS: "积分图" integration graph



∴



= "1" - "2" - "4" + "3"

定义 $F_{X_1,X_2}(x_1,x_2) = P(X_1 \le x_1,\ X_2 \le x_2)$ (Joint Distribution Function)

$\Rightarrow P(X_1 \in [a,b],\ X_2 \in [c,d]) = F_{X_1,X_2}(b,d) + F_{X_1,X_2}(a,c) - F_{X_1,X_2}(a,d) - F_{X_1,X_2}(b,c)$

$P((X_1,X_2) \in A) = \iint_A \frac{\partial^2}{\partial x_1 \partial x_2} F_{X_1,X_2}(x_1,x_2)\,dx_1 dx_2$

$\Rightarrow$ 定义 $f_{X_1,X_2}(x_1,x_2) = \frac{\partial^2}{\partial x_1 \partial x_2} F_{X_1,X_2}(x_1,x_2)$ (Joint Probability Density)

- Boundary Distribution $F_{X_1}(x_1) = P(X_1 \le x_1) = P(X_1 \le x_1,\ X_2 \le \infty)$

$= F_{X_1,X_2}(x_1,\infty) = \int_{-\infty}^{+\infty} \int_{-\infty}^{x_1} f_{X_1,X_2}$

- Boundary Density $f_{X_1}(x_1) = \frac{d}{dx_1} F_{X_1}(x_1) = \frac{d}{dx_1} \int_{-\infty}^{+\infty} \int_{-\infty}^{x_1} f_{X_1,X_2} = \int_{-\infty}^{+\infty} f_{X_1,X_2}(x_1,x_2)\,dx_2$

Boundary $\xleftarrow[?]{\text{Integration}}$ Joint

if independent: $f_{X_1,X_2}(x_1,x_2) = f_{X_1}(x_1) f_{X_2}(x_2)$

$F_{X_1,X_2}(x_1,x_2) = F_{X_1}(x_1) F_{X_2}(x_2)$

$E(g(x_1)h(x_2)) = E(g(x_1))\,E(h(x_2))$

2. $g(x) = h(x) = x$, 则 $E(g(x_1)h(x_2)) = E(x_1x_2)$ correlation 相关

另一种定义: $E(X_1 - EX_1)(X_2 - EX_2) = E(X_1X_2) - EX_1\,EX_2$

if independent, $E(X_1X_2) = E(X)\,EX_2$ uncorrelated

e.g. $\theta \sim U(0, 2\pi)$ $X_1 = \cos\theta$, $X_2 = \sin\theta$ $\because X_1^2 + X_2^2 = 1$ $\therefore$ 不 independent

$E(X_1) = \int_0^{2\pi} \cos\theta \frac{1}{2\pi} d\theta = 0$

$E(X_2) = \int_0^{2\pi} \sin\theta \frac{1}{2\pi} d\theta = 0$

$E(X_1X_2) = \int_0^{2\pi} \cos\theta \sin\theta \frac{1}{2\pi} d\theta = 0$ $\therefore$ uncorrelated

$\therefore$ uncorrelated $\not\Rightarrow$ independent

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3. correlation coefficiant

定义 $corr(X_1, X_2) = \dfrac{E(X_1 - EX_1)(X_2 - EX_2)}{\sqrt{Var(X_1)\ Var(X_2)}} = \dfrac{E(X_1 - EX_1)(X_2 - EX_2)}{\sqrt{E(X_1 - EX_1)^2 E(X_2 - EX_2)^2}}$

命题：$|corr(X_1, X_2)| \le 1$

证，只须证 $E^2(XY) \le E(X^2) E(Y^2)$ (Cauchy - Schwarz)

设 $h(\alpha) = E(\alpha X - Y)^2 = \alpha^2 EX^2 - 2\alpha E(XY) + EY^2 \ge 0$

① $E(X^2) = 0$ ✓ ② $E(X^2) > 0$ $\Rightarrow \Delta = 4E^2(XY) - 4E(X^2)E(Y^2) \le 0$ 证毕

e.g. Mean Square Error

$X_1, \cdots, X_n \overset{i.i.d}{\sim} f_X(x)$

$\left(\frac{1}{n}\right) \sum_{k=1}^{n} (X_k - \underbrace{EX}_{\text{未知}})^2 \longrightarrow \left(\frac{1}{n-1}\right) \sum_{k=1}^{n} (X_k - \bar{X})^2 \qquad \bar{X} = \frac{1}{n} \sum_{k=1}^{n} X_k$

证 $E\left(\frac{1}{n-1} \sum_{k=1}^{n} X_k - \bar{X})^2\right) = Var(X_1)$ 过程略

$\phi_{X_1+X_2}(\omega) = E(\exp(j\omega(X_1 + X_2))) = E\left(\underbrace{\exp(j\omega X_1)}_{g} \underbrace{\exp(j\omega X_2)}_{h}\right)$

$= E(\exp(j\omega X_1))\ E(\exp(j\omega X_2)) = \phi_{X_1}(\omega)\ \phi_{X_2}(\omega)$

L11 Calculate Distribution

1. 引入：$F_X(x) = P(X \le x)$ $\begin{cases} x = g(Y), & Y \sim f_Y \\ x = g(X_1, \cdots, X_n), & (X_1, \cdots, X_n) \sim f_{X_1, \cdots, X_n} \end{cases}$ 求逆即可 / 不好求逆 ⇒ 转化为积分

e.g.1 $X \sim f_X$, $Y \sim f_Y$, $Z = X + Y$ 问：$Z \sim ?$

$F_Z(t) = P(Z \le t) = P(X + Y \le t)$ 设 $A = \{(x, y) \in \mathbb{R}^2 \mid x + y \le t\}$

$= \iint_A f_{X,Y} = \iint_{x+y \le t} f_{X,Y} = \int_{-\infty}^{\infty} \int_{-\infty}^{t-y} f_{X,Y}(x, y)\, dx\, dy$

$f_Z(t) = \frac{d}{dt} F_Z(t) = \int_{-\infty}^{\infty} f_{X,Y}(t-y, y)\, dy$

if independent $\int_{-\infty}^{\infty} f_X(t-y) f_Y(y)\, dy = (f_X * f_Y)(t)$ (convolution)

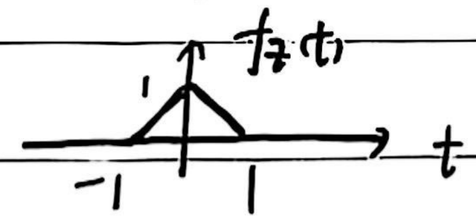
e.g.2 $X, Y \overset{i.i.d}{\sim} U(0,1)$ $Z = X - Y$

类似 e.g.1, $f_Z(t) = \int_{-\infty}^{\infty} f_X(t+y) f_Y(y)\, dy = \int_{-\infty}^{\infty} I_{(0,1)}(t+y) I_{(0,1)}(y)\, dy$

① $t \ge 1$, $f_Z(t) = 0$ ② $t \le -1$, $f_Z(t) = 0$

③ $t \in (0,1)$, $f_Z(t) = \int_0^{1-t} dy = 1 - t$ ④ $t \in (-1, 0]$, $f_Z(t) = \int_{-t}^{1} dy = 1 + t$

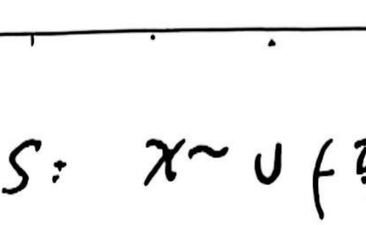
⇒ $f_Z(t) = (1 - |t|)\, I_{(-1,1)}(t)$ 独立均匀分布的和是三角分布



e.g.3 $X, Y \overset{i.i.d}{\sim} U(0,1)$, $\tilde{Z} = \max(X - Y, 0)$

$F_{\tilde{Z}}(t) = P(\max(X-Y, 0) \le t) = P(X - Y \le t, 0 \le t) = \begin{cases} 0 & t < 0 \\ \frac{1}{2} & t = 0 \\ P(X - Y \le t) & t > 0 \end{cases}$

$f_{\tilde{Z}}(t) = \frac{d}{dt} F_{\tilde{Z}}(t) = \begin{cases} 0 & t < 0 \\ \frac{1}{2}\delta(t) & t = 0 \\ f_Z(t) \text{ (e.g.2中的)} & t > 0 \end{cases}$



e.g.4 $X, Y \overset{i.i.d}{\sim} N(0,1)$ $Z = X/Y$, $Z \sim ?$

$F_Z(t) = P(X/Y \le t) = \iint_{X/Y \le t} \frac{1}{2\pi} \exp\left(-\frac{x^2+y^2}{2}\right) dx\, dy$

$= \left(\int_{-\infty}^{0} \int_{ty}^{\infty} + \int_{0}^{\infty} \int_{-\infty}^{ty}\right) \frac{1}{2\pi} \exp\left(-\frac{x^2+y^2}{2}\right) dx\, dy$

$f_Z(t) = \frac{d}{dt} F_Z(t) = \int_{-\infty}^{0} (-y) \frac{1}{2\pi} \exp\left(\frac{-(t^2+1)y^2}{2}\right) dy + \int_{0}^{\infty} y \frac{1}{2\pi} \exp\left(\frac{(t^2+1)y^2}{2}\right) dy$

$= \frac{1}{\pi(1+t^2)}$ (注：此为 Cauchy 分布，Cauchy 分布特点为各阶矩都甲)

(PS: $X \sim U(-\frac{\pi}{2}, \frac{\pi}{2})$, $Z = \tan X$ $Z \sim$ Cauchy 分布)

Date

e.g. (往年题) $X, Y, Z \overset{i.i.d}{\sim} U(0,1)$, $\widetilde{Z} = (XY)^Z$

$$F_{\widetilde{Z}}(t) = P(\widetilde{Z} \le t) = \iiint_{(xy)^z \le t} f_{X,Y,Z}(x,y,z)\,dx\,dy\,dz$$

① 设 $T = XY \in [0,1]$. $(xy)^z \le t \Leftrightarrow T^z \le t \Leftrightarrow z \ge \frac{\ln t}{\ln T}$

$$\Rightarrow \int_{T^z \le t} f_Z(z)\,dz = \begin{cases} \int_{\frac{\ln t}{\ln T}}^{1} dz = 1 - \frac{\ln t}{\ln T} & 0 < T \le t \le 1 \\ 0 & 0 < t < T \le 1 \\ 1 & t > 1 \end{cases}$$

② 从而, $F_{\widetilde{Z}}(t) = \iint_{T \le t} \left(1 - \frac{\ln t}{\ln T}\right) f_T(T)\,dT$ $\quad (0 < T \le t < 1)$

$$F_{XY}(T) = P(XY \le T) = \iint_{xy \le T} I_{(0,1)}(x)\, I_{(0,1)}(y)\,dx\,dy$$

$$= \left(\int_0^T \int_0^1 + \int_T^1 \int_0^{\frac{T}{y}}\right) I_{(0,1)}(x)\, I_{(0,1)}(y)\,dx\,dy$$

$$= \int_0^T dy + \int_T^1 \frac{T}{y}\,dy = T - T\ln T$$

$$\Rightarrow f_T(T) = -\ln T$$

代入, $F_{\widetilde{Z}}(t) = \int_0^t \left(1 - \frac{\ln t}{\ln T}\right)(-\ln T)\,dT = t \quad (0 < t \le 1)$

$$\Rightarrow (XY)^Z \sim U(0,1)$$

2. $X, Y \overset{i.i.d}{\sim} N(0,1)$, $\begin{cases} R = (X^2+Y^2)^{\frac{1}{2}} \\ \Theta = \tan^{-1}\left(\frac{Y}{X}\right) \end{cases}$ 问: $R, \Theta \sim ?$

思路: $P((R,\Theta) \in A) = \iint_A f_{R,\Theta}(r,\theta)\,dr\,d\theta$

Ⅱ

$$P((X,Y) \in A') = \iint_{A'} f_{X,Y}(x,y)\,dx\,dy \overset{\text{积分换元}}{=\!=\!=} \iint_A f_{X,Y}(x(r,\theta), y(r,\theta))\,|J|\,dr\,d\theta$$

本题: $\iint_{\mathbb{R}^2} \frac{1}{2\pi} \exp\left(-\frac{x^2+y^2}{2}\right)dx\,dy = \iint_{[0,+\infty)\times[0,2\pi)} \boxed{\frac{1}{2\pi}\exp\left(-\frac{r^2}{2}\right) r}\,dr\,d\theta$

$$f_{R,\Theta} = r\exp\left(-\frac{r^2}{2}\right) I_{(0,\infty)}(r) \cdot \frac{1}{2\pi} I_{[0,2\pi)}(\theta)$$

总结:
- 一对一 : 求逆
- 多对一 : 积分
- n 对 n : 积分换元
- n 对 m ($m<n$): 先积分换元, 再取 boundary

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L12 Conditional Expectation

1. X r.v., $f_X(x)$, $E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx$

$f_X(x) \to f_{X|Y}(x|y) = \dfrac{f_{X,Y}(x|y)}{f_Y(y)}$ } misleading

$E(X|Y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx$

正确理解：X, Y r.v.，现在定义第三个 r.v. $E(X|Y)$ 消除的是 X 的随机性，还剩 Y 的随机性

① $E(E(X|Y)) = E(X)$

$E(g(X,Y)) = E_Y\left(E_{X|Y}(g(X,Y)|Y)\right)$ 条件住了 Y，Y 的随机性暂时消失，计算 X

② $E(X h(Y)|Y) = h(Y)\,E(X|Y)$ 由于 Y 的随机性暂时消失 ∴ 可以提出来

e.g.1 $X_1, \cdots, X_n$ i.d.，则 $E(X_1+\cdots+X_n) = E(X_1)+\cdots+E(X_n) = n\,E(X_1)$

if N r.v. independent of $\{X_k\}$，则 $E(X_1+\cdots+X_N) = E(N)\,E(X_1)$

证：$E(X_1+\cdots+X_N) = E_N\left(E_{X|N}(X_1+\cdots+X_N|N)\right)$ 先条件住一个 r.v.，处理另一个，然后再处理剩下条件住的那个 $\overset{X|N\sim X}{=\!=\!=} E(N\,E(X_1)) = E(N)\,E(X_1)$

e.g.2 X, Y r.v., $g(X) \to Y$, $\min_g E_{X,Y}(Y-g(X))^2$ $\Rightarrow g_{opt}(X) = E(Y|X)$

证：$E_{X,Y}(Y-g(X))^2 = E(Y-E(Y|X)+E(Y|X)-g(X))^2$

$= E(Y-E(Y|X))^2 + E(E(Y|X)-g(X))^2 + 2E(Y-E(Y|X))(E(Y|X)-g(X))$

目标 ≥ 0 只须证 $=0$

考察 $E(Y-E(Y|X))(E(Y|X)-g(X))$

$Y \quad X \quad X \quad X$ $\Rightarrow$ 条件住 X

$= E_X\left(E_{Y|X}\left((Y-E(Y|X))(E(Y|X)-g(X))\,|X\right)\right)$

无随机性了，提出来 $E(Y|X)E(1|X) = E(Y|X)$

$= E_X\left[(E(Y|X)-g(X))\,E_{Y|X}\left((Y-E(Y|X))|X\right)\right] = E(Y|X) - E(E(Y|X)|X) = 0$

无随机性了，提出来

证毕

2. Conditional Variation

$Var(X) = E(X-E(X))^2$

$Var(X|Y) = E((X-E(X|Y))^2|Y)$ 有随机性，Var 也是 r.v.

$Var(X) = E(Var(X|Y)) + Var(E(X|Y))$ (证法 = 上，略)

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3. 例题 (" First Step Analysis)

e.g.1 Geometrical Distribution $P(N=k) = (1-p)^{k-1}p$ $E(N) = \frac{1}{p}$

∵ N 不独立于 X ∴ $E(X_1 + \cdots + X_N)$ ~~条件性N $E(N)E(X_1)$~~

⇒ 条件性X $E(N) = E\left(E(N|X_1)\right)$

$E(N) = p \cdot \underbrace{E(N|X_1=1)}_{=1} + (1-p) \cdot \underbrace{E(N|X_1 \neq 1)}_{=1+E(N)}$ ⇒ $E(N) = \frac{1}{p}$

e.g.2 Dice "11"

$E(N) = E\left(E(N|X_1)\right) = p \cdot E(N|X_1=1) + (1-p) \cdot E(N|X_1 \neq 1)$

$= p\left[p \cdot \underbrace{E(N|X_1=1, X_2=1)}_{=2} + (1-p) \cdot \underbrace{E(N|X_1=1, X_2 \neq 1)}_{=2+E(N)}\right] + (1-p)(1+E(N))$

⇒ $E(N) = \frac{1+p}{p^2}$

e.g.3 Coin "10"

$E(N) = p \cdot E(N|X_1=1) + (1-p) \cdot E(N|X_1=0)$

$= p \cdot (1 + \underbrace{E(G)}_{\text{几何 (∵只有0和1)}}) + (1-p) \cdot (1+E(N))$

(先1后0) (先1后1)

PS: "10" 比 "11" 更容易出现 ∵ "11" 若扔出了0，则前面白费；而 "10" 不会

e.g.4 (考题) ① Dice "526"

$E(N) = (1-p) \cdot \underbrace{E(N|X_1 \neq 5)}_{=1+E(N)} + p \cdot E(N|X_1=5)$

$E(N|X_1=5) = p \underbrace{E(N|X_1=5, X_2=2)}_{=2} + p \underbrace{E(N|X_1=5, X_2=5)}_{=1+E(N|X_1=5)} + (1-2p) \underbrace{E(N|X_1=5, X_1 \neq 5, \neq 2)}_{=2+E(N)}$

⇒ $E(N) = \cdots$

② "526" ③ "526526"

一. Limit in Probability Theory

1. 考察 $X_n \xrightarrow{\text{收敛于}} X$ $\quad X: \Omega \to \mathbb{R}$

回顾：① Pointwise：$f_n(x) \to f(x)$：$\forall x \in D$, $\forall \varepsilon > 0$, $\exists N_x \in \mathbb{N}$, s.t. $\forall n > N$, $|f_n(x) - f(x)|$

② Uniformly：$f_n(x) \to f(x)$：$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, $\forall x$, $\forall n > N$, $|f_n(x) - f(x)| \le \varepsilon$

③ m.s.（均方误差）：$f_n(x) \to f(x)$：$\int_D |f_n(x) - f(x)|^2 dx \to 0$ 或a

⇒ Probability：① $P(\omega \in \Omega \mid \lim_{n\to\infty} X_n(\omega) = X(\omega)) = 1$ "Almost Everywhere"（记作 $X_n \xrightarrow{a.}$

（不要求每个点都收敛，只要求不收敛的点根测度为零）

② $\forall \varepsilon > 0$, $P(\omega \in \Omega \mid |X_n(\omega) - X(\omega)| \ge \varepsilon) \xrightarrow{n\to\infty} 0$ In Probability

③ $F_{X_n}(x) \xrightarrow{n\to\infty} F(x)$ at continous point of $F_X(x)$ In Distributio

e.g. $X \sim N(0,1)$, $X_n = (-1)^n X$ ① X ② X ③ $X \sim N(0,1)$, $-X \sim N(0,1)$ ∴ ✓

命题：① ⇒ ② ⇒ ③ 证明略

2. Limit Theorem：Law of Large Numbers

$\{X_n\}$ $\xrightarrow{\frac{X_1 + \cdots + X_n}{n}}$ $\xrightarrow{n\to\infty}$ $E(X)$ （设 $X_1, \cdots, X_n \overset{i.i.d}{\sim} f_X(x)$）

两种收敛方式：Strong Law："Almost Everywhere"

Weak Law："In Probability"

"Concentration" $P(|X - EX| \ge \varepsilon) \le \frac{Var(X)}{\varepsilon^2}$

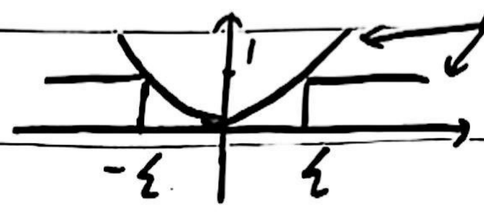
⇑

Chebyshev Inquality：$P(|X| \ge \varepsilon) \le \frac{E|X|^2}{\varepsilon^2}$

「证：法一) $E|X|^2 = \int_{\mathbb{R}} x^2 f_X(x)\,dx \ge \int_{|x|\ge\varepsilon} x^2 f_X(x)\,dx \ge \int_{|x|\ge\varepsilon} \varepsilon^2 f_X(x) = \varepsilon^2 \int_{|x|\ge\varepsilon} f_X(x)$

法二) 把 P 和 E 联系起来（$P(A) = E(I_A)$，概率是期望的 Indicator） $= \varepsilon^2 P(|X| \ge \varepsilon)$

即证 $P(|X| \ge \varepsilon) = E\left(I_{\{|X(\omega)| \ge \varepsilon\}}(\omega)\right) \le E\left|\frac{X}{\varepsilon}\right|^2$



由图 证毕 」

证：切比雪夫版本 Weak Law 略

Date

3. Central Limit Theorem (中心极限Thm)

设 $\{X_n\}$ i.i.d $E(X_n)=0$ $Var(X)=1$, 则

$$\frac{X_1+\cdots+X_n}{\sqrt{n}} \xrightarrow[\text{in distribution}]{n\to\infty} N(0,1)$$

证(用 Characteristic Function):

$$\phi_X(w) = E\left(\exp(jwX)\right) = \int_{\mathbb{R}} f_X(x)\exp(jwx)\,dx$$

$$\phi_{\frac{X_1+\cdots+X_n}{\sqrt{n}}}(w) = E\left(\exp\left(j\tfrac{w}{\sqrt{n}}(X_1+\cdots+X_n)\right)\right) = \left[E\left(\exp\left(j\tfrac{w}{\sqrt{n}}X_1\right)\right)\right]^n$$

$$= \left(\phi_{X_n}\left(\tfrac{w}{\sqrt{n}}\right)\right)^n = \left[1+j\tfrac{w}{\sqrt{n}}E(X_1)+\tfrac{1}{2}E\left(j\tfrac{w}{\sqrt{n}}X_1\right)^2+o(\tfrac{1}{n})\right]^n$$

$$= \left[1-\tfrac{1}{2}\tfrac{w^2}{n}+o(\tfrac{1}{n})\right]^n \xrightarrow{n\to\infty} \exp\left(-\tfrac{w^2}{2}\right) \quad \text{Gauss!}$$

总结: $\frac{X_1+\cdots+X_n}{n}$ 独立 r.v. 相加, 随机性个 但被除以的 n 抵消掉了

$\frac{X_1+\cdots+X_n}{\sqrt{n}}$ $\sqrt{n}$ 不如 n 强. ∴ 没把随机性完全平滑掉; 但个性都被抹掉. 只有共性随机性

⇒ n: 抵消没了

$\sqrt{n}$: 还剩 Gauss

临界点: $\sqrt{n\ln\ln n}$!